

SIMATS
ENGINEERING



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Saveetha Institute of Medical And Technical Sciences
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Statistical Learning-Based Weather Prediction

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Introduction

- Weather prediction forecasts future atmospheric conditions.
- Statistical Learning learns patterns from historical weather observations.
- Important variables include temperature, humidity, pressure, wind speed, and rainfall.
- The learned relationships can be used to predict future weather conditions.

Problem Statement and Objectives

Problem Statement

- Weather conditions change continuously and depend on multiple factors.
- Manual analysis of large historical weather records is difficult.

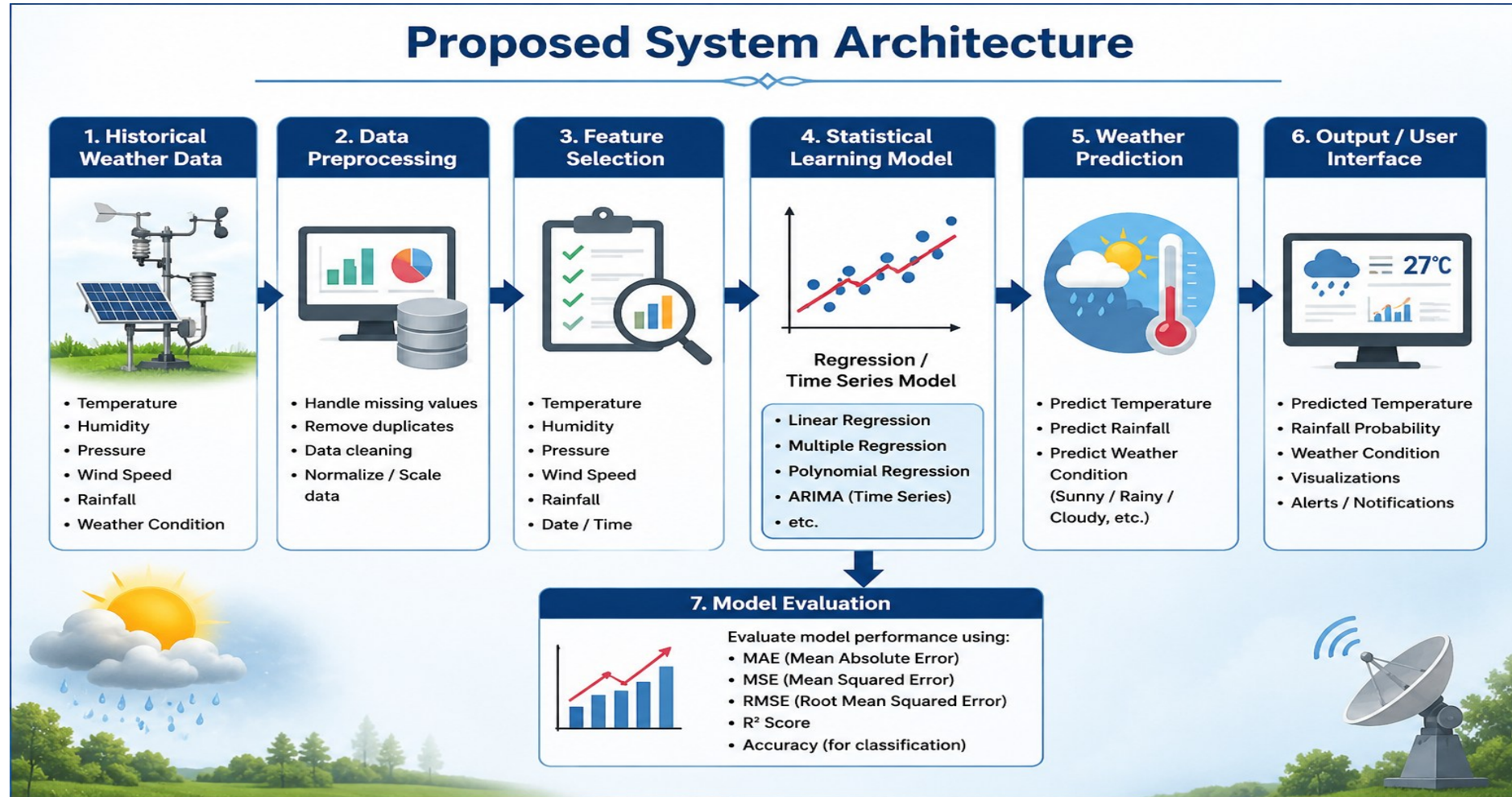
Objectives

- Collect historical weather data.
- Identify relationships between weather variables.
- Train a statistical learning model.
- Predict future weather conditions.
- Evaluate prediction performance.

Weather Dataset and Features

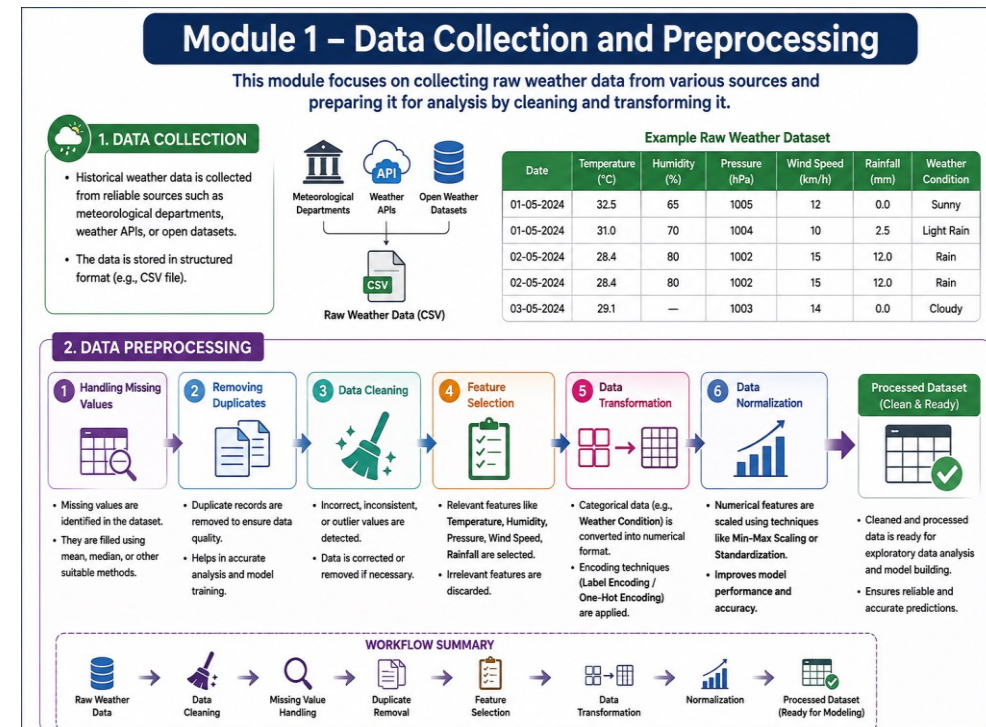
Day	Temperature	Humidity	Wind Speed	Rainfall	Weather
1	30°C	70%	12 km/h	0 mm	Sunny
2	29°C	75%	15 km/h	2 mm	Cloudy
3	27°C	82%	18 km/h	8 mm	Rainy
4	31°C	65%	10 km/h	0 mm	Sunny
5	28°C	80%	20 km/h	6 mm	Rainy

Proposed System Architecture



Module 1 – Data Collection and Preprocessing

- Collect historical weather observations.
- Handle missing and inconsistent values.
- Remove duplicate records.
- Convert data into suitable numerical form.
- Select relevant weather features.
- Split data into training and testing sets.
- Output: Clean weather dataset.



Module 2 – Statistical Learning

- Statistical Learning uses data and statistical techniques to learn relationships between variables.
- It estimates patterns from observed weather data.
- The learned model is applied to unseen or future observations.
- Example: temperature, humidity, and pressure can be used to estimate tomorrow's temperature.

Module 3 – Weather Prediction

- INPUT: Temperature, Humidity, Pressure, Wind Speed, Rainfall
- MODEL: Trained Statistical Learning Model
- OUTPUT: Predicted Temperature / Weather Condition

Sample Prediction

- Temperature: 28°C
- Humidity: 80%
- Pressure: Low
- Wind Speed: 20 km/h
- Illustrative prediction: Rainy weather
- The model uses patterns learned from historical observations.

Module 4 – Model Evaluation & Performance

- Test the trained model using unseen weather data.
- For continuous prediction: MAE, MSE, and RMSE.
- For classification: Accuracy, Precision, Recall, and F1-Score.
- Lower prediction error indicates better continuous prediction performance.

Advantages and Applications

Advantages

- **Uses historical data** to identify weather patterns.
- **Handles multiple weather factors** such as temperature, humidity, pressure, wind speed, and rainfall.
- **Provides fast predictions** once the model is trained.
- **Improves decision-making** by providing data-driven forecasts.
- **Can be updated** when new weather data becomes available.

Applications

-  **Rainfall Prediction** – Predict the possibility and amount of rainfall.
-  **Temperature Forecasting** – Predict future temperature.
-  **Storm Prediction** – Identify possible severe weather conditions.
-  **Agricultural Planning** – Help farmers plan irrigation and crop activities.

Conclusion

- Historical weather data is collected and preprocessed.
- Statistical Learning identifies useful relationships.
- The trained model predicts future weather conditions.
- Performance is evaluated using suitable statistical metrics.
- It supports faster, data-driven weather forecasting.

References

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Thank
you!